Supplementary Experimental Report: Raw-Frame Analysis, Anchor Lower-Trim Experiment, and Phase Center Sensitivity

Erlangen 28-May-2026 V5 Campaign – Additional Analysis

Zekai Xiao
OLAO Health GmbH / LMU Muenchen

June 2026

Contents

1 Context

This document supplements the original English Vicon validation report and records the experiments that were completed after that report was sent for review. The original campaign left a central paradox: the V5 common-mode anchor layout corrected the anchor-side metric scale, with Sim3 scale moving from 0.958 for V4 to 1.010 for V5, yet the older V4 layout still gave the lower p50 static median on the same 24-position Erlangen campaign. Under the unified scalar convention, V4+C_V4+D_LOO gives 61.5 mm median 3D error, while V5+C_V5+D_LOO gives 63.5 mm. The gap is only 2.0 mm, so the baseline paradox is narrower than the earlier mixed-convention table implied. These values are reproduced from `FULL_V5_convention_unification/tables/unified_headline_table.csv`.

The additional experiments were designed to test whether this paradox is caused by delay–layout coupling rather than by a simple V4/V5 implementation error. The phase-center sensitivity analysis asks whether Vicon marker offsets can overturn the ranking. The raw-frame campaign uses the stored per-frame tag-anchor distributions to reduce positive NLOS tails. The anchor lower-trim blind experiment asks whether the same lower-tail statistic should also be used for inter-anchor self-calibration. The solver audit records which parts of the V1–V5 and T1–T4 solvers are actually algorithmic changes and which parts are preprocessing or policy changes.

The main conclusion is narrower and stronger than an accuracy claim. V5 is the physically better anchor geometry by metric scale, while V4 is only the marginal p50 single-environment positioning winner. When the positive tag-anchor range tail is reduced in static batch mode, the empirical ranking flips: V5 reaches 44.5 mm LOO, whereas the best V4 geometry row in the same raw-frame ranking is 53.9 mm. The baseline gap is 2.0 mm, but the intervention gives V5 a 9.4 mm advantage, which is the strongest causal evidence that V4's p50 advantage came from beneficial cancellation between a distorted layout and structured positive range bias.

2 Phase Center Sensitivity Analysis

The phase-center sensitivity batch addressed a necessary alternative explanation. Vicon marker centers are not guaranteed to coincide with UWB antenna phase centers, and a systematic marker-to-antenna offset could make a geometrically correct UWB layout appear worse against the optical reference. The analysis therefore perturbed phase centers and repeated the relevant ranking checks rather than assuming that the marker geometry was exact.

The global shift sweep found that the V4-over-V5 p50 static ranking does not flip for tested offsets up to more than 10 mm. The manufacturing-variation Monte Carlo used 5000 samples per sigma level and found that $P(\text{V4 beats V5})$ remains at least 0.998 through $\sigma = 8$ mm. The Vicon oracle rank is less stable: the table reports a fragile threshold around 2 mm for the oracle rank or worst-status conclusion. The cancellation valley itself remained invariant to the tested phase-center offsets, which means that the delay–layout coupling result is not an artifact of one exact optical marker model.

Table 1: Phase-center robustness summary. Values are reproduced from `FULL_V5_phase_center_sensitivity/tables/a6_robustness_summary.csv`.

Conclusion	Baseline value	Flip threshold	Robustness
V5 Sim3 scale > 0.99	1.010	>10	robust
V4+LOO beats V5+LOO	V4-V5=-2.0 mm under scalar convention	>10 in prior sweep	robust ranking, small margin
Vicon oracle rank/worst status	rank=2, worst=False	2.0	fragile
D_tag LOO approximately 49.6mm	49.028 mm; sensitivity 0.190 mm/mm	not binary	stable
D_tag per-height spread V5 < V4	7.4 < 11.8 mm from prior mechanism audit	not directly flipped by global phase-center sweep	not directly tested here; use A4 as caveat
Cancellation valley exists	max tested operating-point valley-distance shift 11.37	does not depend on absolute phase-center offset	invariant mechanism

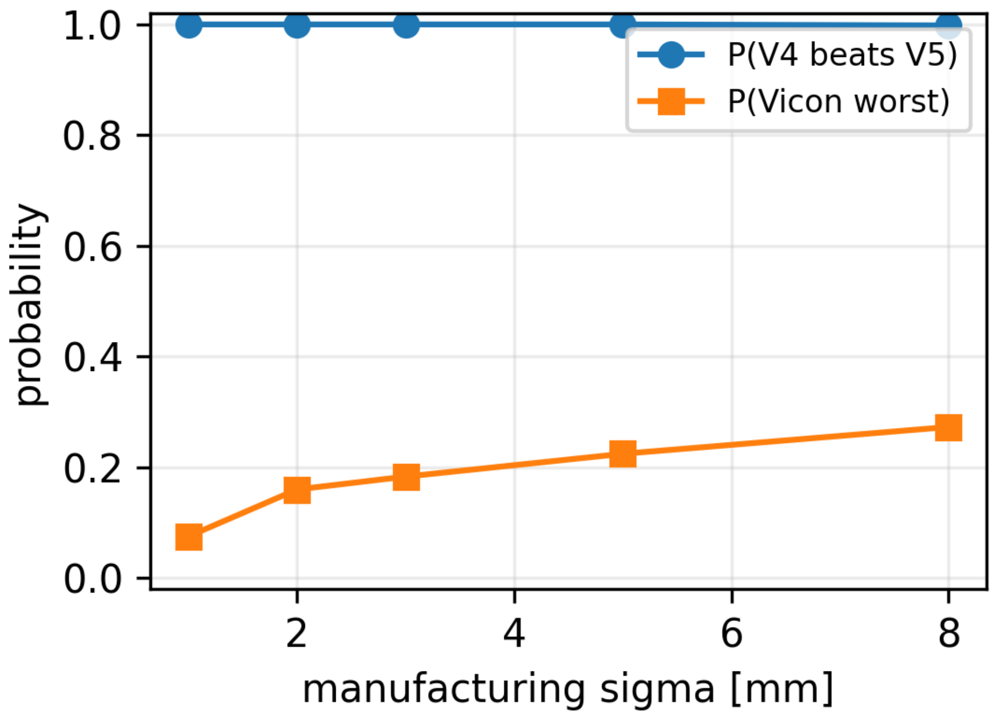

Figure 1: Manufacturing-variation phase-center Monte Carlo. The static V4-over-V5 p50 ranking is robust across the tested perturbation range, while the Vicon oracle rank is more fragile. Source: `FULL_V5_phase_center_sensitivity/figures/a2_ranking_probability_vs_sigma.png`.

3 Raw-Frame Range Distribution Analysis

The raw-frame campaign began with a data-availability check. The static captures contain the raw per-frame tag-anchor observations needed for distribution-level analysis: 230,544 raw rows, 228,265 valid rows, and approximately 1200 frames per position-anchor link. This matters because the historical p50 pipeline collapses each link histogram to one median value before solving. If the histogram has a positive NLOS tail, a lower-quantile statistic can move the measurement closer to the LOS-side component without using CIR data.

Table 2: Raw-frame campaign key numbers. Values are from `FULL_V5_experimental_report/tables/rawframe_v3_key_results.csv`.

Metric	Value	Unit
Raw rows	230544.0	rows
Valid rows	228265.0	rows
Full grid configurations	107448.0	rows
Best honest LOO median	44.5	mm
Oracle lower bound	44.6	mm
Bootstrap CI	[33.4, 82.8]	mm

The important result is the ranking flip. Under scalar p50 aggregation, V4 barely wins the static LOO comparison despite having the wrong metric scale. Under lower-quantile static batch aggregation, V5 becomes the best geometry: lower_trim_20 with V5 and Huber30 gives 44.5 mm median LOO, while the best V4 geometry row in the same honest raw-frame ranking is 53.9 mm and the best Vicon geometry row is 44.7 mm. The oracle lower bound in the v3 ladder is 44.6 mm, so the V5 lower-tail row nearly reaches the recoverable median implied by the available raw distributions.

This is not presented as a new NLOS detection method. The lower-tail statistic is deliberately simple. Its value is experimental: it reduces the positive tag-anchor tail enough to remove the cancellation condition under which V4 was winning. The intervention therefore supports the mechanism claim that single-environment positioning accuracy can prefer a physically wrong geometry when the wrong geometry cancels structured range bias.

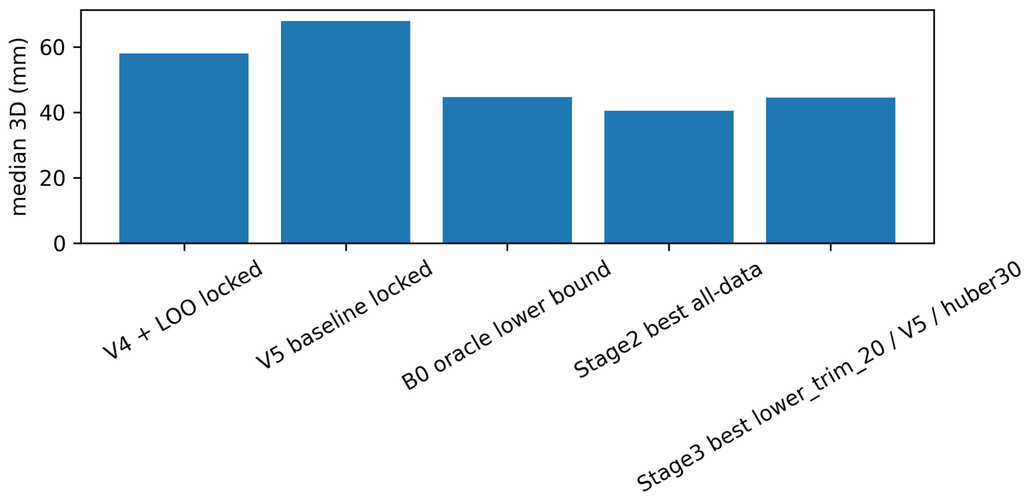

Figure 2: Raw-frame accuracy ladder. The lower-quantile V5 static batch row approaches the oracle median, while the tail remains unresolved. Source: `FULL_V5_rawframe_bruteforce_v3/figures/s6_accuracy_ladder.png`.

The caveat is the tail. Under the unified scalar convention, the lower-quantile V5 row improves the median to 44.5 mm and improves P95 from 175.1 mm for V5 scalar p50 to 164.1 mm, with RMSE 81.5 mm. This removes the old apparent V4 P95 advantage, which came from mixing evaluation conventions. The tail is still high in absolute terms, so the lower-tail result solves typical static error better than it solves worst-case static positioning. The 5000-iteration bootstrap interval, 33.4–82.8 mm, also remains wide because the campaign has only 24 static positions.

4 Anchor Lower-Trim Blind Experiment

The raw-frame tag-anchor result raised a second question: if lower-tail statistics recover a better LOS-side tag-anchor range, should the same statistic be used for inter-anchor self-calibration? The blind anchor lower-trim experiment tested this directly by rebuilding V5-style anchor layouts with several inter-anchor aggregation methods and e_i settings, then replaying the tag positioning pipeline.

The answer was negative. The inter-anchor distributions contain 56,000 valid rows across 28 pairs, with a median of 2000 frames per pair. Their mean skewness is 0.063, which is close to symmetric, while the mean p50 minus lower_trim_20 difference is 32.9 mm. In other words, lower_trim_20 systematically shortens the inter-anchor distances even though those distributions do not show the same right-tail structure as the static tag-anchor histograms.

Table 3: Anchor lower-trim blind-experiment key numbers. Values are from `FULL_V5_experimental_report/tables/anchor_lower_trim_key_results.csv`.

Metric	Value	Unit
AA valid rows	56000.0	rows
Median frames per pair	2000.0	frames
Mean skewness	0.063	skewness
Mean p50 - lower_trim_20	32.9	mm
Best lower_trim_20 anchor row	46.4	mm LOO
Current p50 control	44.5	mm LOO
Best p50 e_i=0 row	43.2	mm LOO
P(new wins), e_i=0	0.659	probability

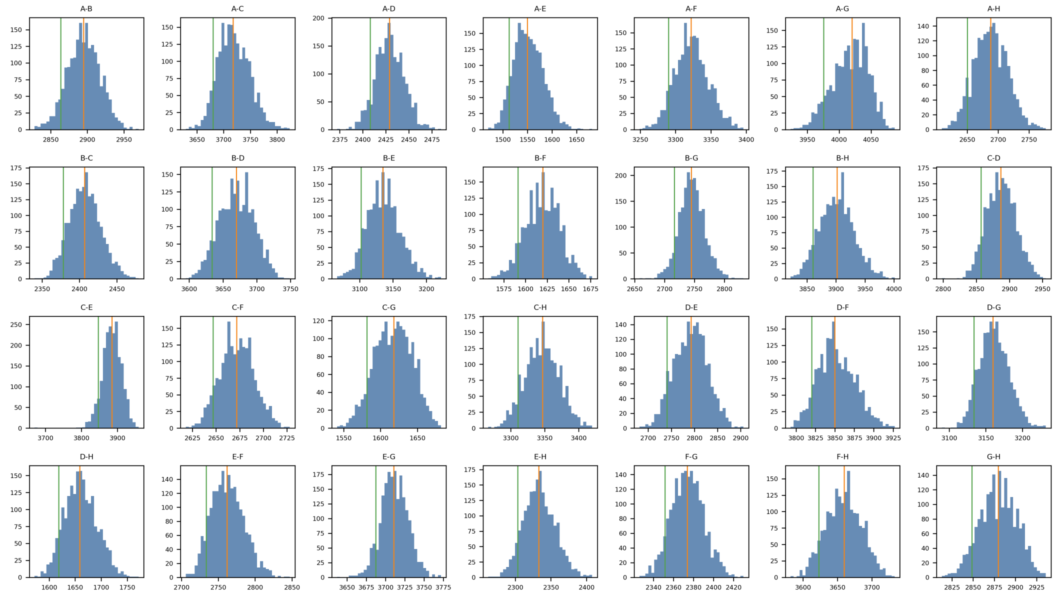

Figure 3: Inter-anchor raw range histograms. The near-symmetric distributions explain why lower-tail inter-anchor aggregation introduces a downward range bias rather than removing a positive tail. Source: `FULL_V5_anchor_lower_trim/figures/11_inter_anchor_histograms.png`.

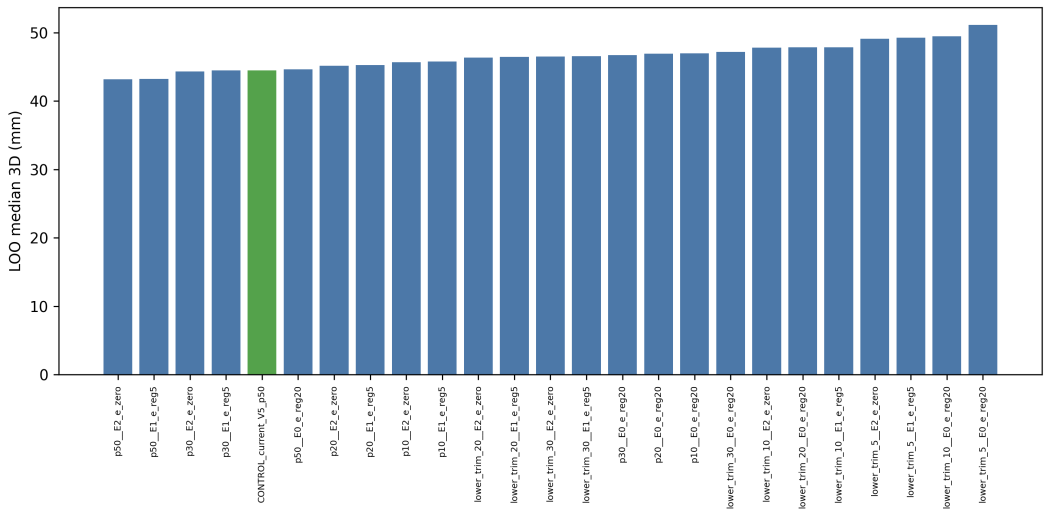

Figure 4: Blind anchor-aggregation comparison. The best lower_trim_20-anchor row is worse than the p50 control, while p50 with e_i=0 is a small, not yet statistically locked, candidate improvement. Source: `FULL_V5_anchor_lower_trim/figures/13_accuracy_by_anchor_method.png`.

The practical rule from this experiment is simple: keep p50/median for inter-anchor self-calibration in this dataset, and treat lower-quantile aggregation as a tag-anchor static batch estimator. The e_i result is separate. p50 with e_i=0 reaches 43.2 mm LOO, compared with 44.5 mm for the current p50/e_reg20 control, but the paired bootstrap gives $P(\text{new wins}) = 0.659$ and a confidence interval crossing zero. This is a candidate setting for the next capture, not a locked replacement.

5 Active Vertical-Geometry Injection Control (RotArm)

The vertical component of the anchor calibration error invites a geometric reading, but it does not survive inspection. The eight anchors form a non-coplanar two-layer, approximately cuboid array with well-conditioned vertical geometry, so vertical observability for this standard layout is not in question and no vertical Fisher-information deficit is expected from it. The residual vertical error therefore cannot be an observability problem and must be systematic. The open question is whether this systematic vertical error is reducible by supplying additional, vertically-diverse calibration measurements, or whether it is a fixed per-device bias. This section settles that question with a direct intervention that floods the self-calibration with strong vertical excitation.

The intervention uses a rigid arm carrying two UWB tags at fixed radii of 426 and 554 mm on opposite sides of the rotation centre. Seventeen captures were recorded with the rotation plane tilted from horizontal (about 1°) to vertical (about 90°) in roughly five levels, so that a tilted rotation contributes vertical excursion $2r \sin \alpha$, and therefore anisotropic vertical information, to the joint anchor solve. If the systematic vertical error were reducible by added vertical calibration excitation, vertical accuracy should improve monotonically as the rotation plane, and hence the injected vertical excursion, approaches vertical.

Arm rigidity and reference quality were verified from the motion-capture markers before use. The inter-tag distance is rigid to about 0.3–0.7 mm along the arm, while the out-of-plane wobble is small and grows with tilt, from about 1 mm near horizontal to about 2.8 mm near vertical; this is far below the ranging noise and is modelled as a soft out-of-plane compliance. The optical wand markers are mislabelled in roughly half of all frames, where the inter-tag distance collapses from about 990 mm to near zero, so all rigidity and tilt quantities use only the clean cluster, and the rotation tilt used as the independent variable is taken from motion capture rather than from the solver.

Each capture was solved against the inter-anchor sweep and the 24 static positions, with and without the rotation injection, using two independent estimators: a free-point solver that treats every tag frame as an unconstrained 3D point, and a full rigid-arm solver that collapses each frame to a single rotation-angle unknown and so forces range residuals onto the anchor geometry. The solved layouts were aligned to the motion-capture anchor reference, and the vertical component of the anchor error was recorded.

The result is a clear negative, summarised in Table 4. Across all sixteen analysable captures the rigid-arm vertical anchor error shows no negative dependence on rotation tilt; the fitted slope is +0.03 mm/deg with correlation $r = 0.08$, that is, essentially flat with the wrong sign. Every single-injection condition is worse than the no-injection baseline of 68.8 mm, and the near-vertical condition, where the injected vertical information is largest, is the worst. Combining all seventeen rotations improves the vertical error only marginally and tilt-independently, to 63.4 mm, consistent with a generic increase in tag-to-anchor constraints rather than vertical-information injection, and it does not reach the no-injection production layout at 59.8 mm. The free-point estimator gives the same conclusion, so the result is not an artefact of one model.

Table 4: RotArm vertical-injection dose–response, binned by motion-capture rotation tilt. Values reproduced from `literature_search/rotarm_zinjection_rigid_results.csv`.

Condition	n	Rigid vertical error (mm)	vs baseline (mm)
Baseline, no injection	–	68.8	–
Tilt $\approx 1°$	1	83.7	+14.9
Tilt $\approx 22°$	4	76.8	+8.0
Tilt $\approx 48°$	4	81.1	+12.3
Tilt $\approx 72°$	4	72.4	+3.6
Tilt $\approx 90°$ (vertical)	3	86.9	+18.1
All rotations combined	17	63.4	−5.4
Production layout, no injection	–	59.8	−9.0

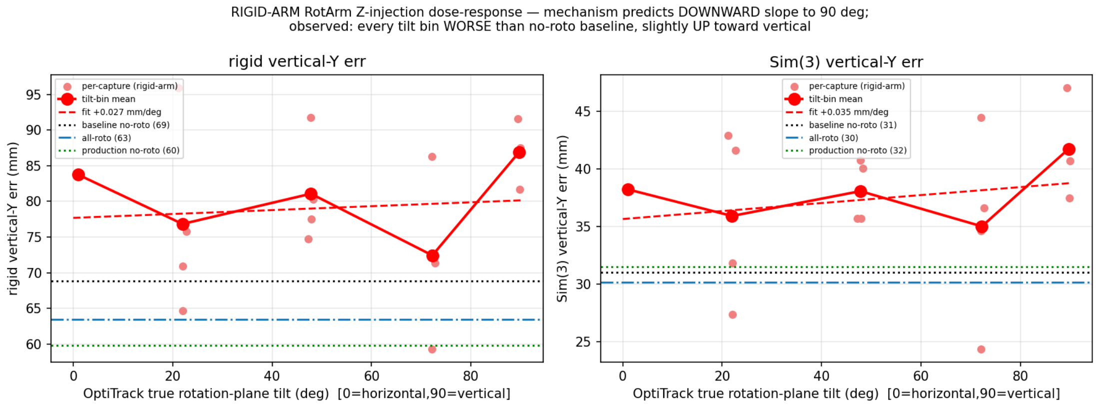

Figure 5: Rigid-arm vertical-injection dose–response against motion-capture rotation tilt. The mechanism predicts a downward slope toward 90°; the observed dependence is flat and slightly positive, every tilt bin is worse than the no-injection baseline, and the near-vertical bin is the worst. Source: `literature_search/rotarm_zinjection_rigid_dose_response.png`.

The result confirms that the systematic vertical error is not reducible by richer vertical calibration data. Actively injecting strong vertical excitation does not reduce the vertical error and, if anything, slightly increases it, whereas tag-delay correction reduces the vertical static error from 61.9 mm to 39.3 mm. This is exactly what is expected when the layout's vertical observability is already good: there is no information deficit for the injection to fill, so the residual vertical bias is the vertical projection of the per-device range-bias and antenna-delay coupling. The value of this result is as a confirmation that closes a data-coverage alternative, not as the discovery of a new mechanism. The absolute error levels here come from a re-analysis pipeline whose baseline, 68.8 mm, is slightly worse than the production layout at 59.8 mm, so only the internal with-and-without and tilt-dose comparisons are used as evidence.

6 Solver Audit Summary

The solver audit found five core anchor-layout solver generations and four C-level tag solver versions, plus one Python tag-solver policy variant. The audit matters because several campaign results are preprocessing or parameterization effects rather than new numerical solvers. In particular, no production T5 tag solver was found; the lower-trim and Huber30 improvements are changes to range aggregation and loss configuration around the existing tag-solver family.

Table 5: Anchor solver evolution from the read-only audit. Source: `../../../../Analysis/solver_audit/reports/SOLVER_AUDIT_SUMMARY.md`.

Version	One-line description	Delay model or key parameter
V1 / v1-old	early MDS baseline	no delay
V2	inverse-variance bidirectional fusion	no delay
V3-lite	median/MAD/MVUE fusion	no delay
V3-full	robust fusion plus alternating per-anchor delays	per-anchor delays
V4 / v4-io	bounded independent-delay Huber inter-anchor solve	dA=0, [-60,+60] mm
V5 / common-mode	common-mode extension, di=c+ei	c=111.985 mm, e_reg=20 mm artifact

Table 6: Tag solver evolution from the read-only audit. Source: `../../../../Analysis/solver_audit/reports/SOLVER_AUDIT_SUMMARY.md`.

Version	One-line description	Status
T1	robust WLS multilateration	current method default
T2	T1 plus quality-aware sigma inflation	C method 2
T3	T2 plus residual EMA and temporal prior	C method 3
T4	adaptive full-anchor T1 / low-redundancy T3 policy	Python wrapper plus C method 4
T4_V6_IMU_GATE	T4 plus accelerometer-gated temporal prior	policy variant

The current research configuration is V5 common-mode self-calibration with the existing artifact at $c = 111.985$ mm and e_reg20, combined with a tag solver that defaults to robust WLS and Huber scale 30 mm after recent edits. The audit recommends keeping p50 inter-anchor aggregation, testing lower-tail tag-side aggregation for static batch operation, explicitly labeling e_i settings, and fixing the C-core loss-selector mismatch where the Python interface exposes a linear option but C validation resets it to Huber.

7 Updated Headline Table

The headline table below is reproduced from `FULL_V5_convention_unification/tables/unified_headline_t`
It should be read as a campaign registry rather than a single leaderboard, because rows differ in evaluation protocol. Rows labelled in-sample, diagnostic, Sim3, or oracle should not be used as deployment claims. Row O is the new raw-frame static batch result; its median is strong, and under the scalar convention its P95 is slightly better than the V5 p50 baseline, although still high in absolute terms.

Table 7: Updated locked headline table. Empty cells indicate metrics not reported for that row in the source table.

Row	Variant	Median	P95	RMSE	Evaluation	New
A	V4 production	81.6	189.4	114.3	in-sample, all 24	
B	V4 + D_LOO	61.5	157.0	87.5	LOO-CV	
C	V5 baseline	63.5	175.1	91.8	LOO-CV	
D	V5 apparent best	65.5	132.0	84.9	in-sample post-selected	
E	V4 apparent best	57.1	221.6	94.4	in-sample post-selected	
F	V5 corrected	75.1			OOB-bootstrap	
G	V4 corrected	66.7			OOB-bootstrap	
H	V5 bootstrap CI	[54.3, 63.7]			bootstrap 95% CI	
I	Nested CV (height)	82.9			held-out test	
J	Nested CV (quadrant)	88.0			held-out test	
K	Nested CV (spatial6)	94.2			held-out test	
L	ROTO V5 per-frame	101.5	214.4	126.2	BEST-FIT-ALIGNED	
M	ROTO SE(3) aligned	82.5	185.2	103.7	diagnostic	
N	ROTO Sim3 aligned	74.3	160.8	94.8	diagnostic only	
O	lower_trim_20 + Huber30 + V5	44.5	164.1	81.5	LOO-CV	YES
P	lower_trim_20 + Huber30 + V5(e_i=0 anchor refit)	43.2	163.1	81.8	LOO-CV; anchor refit diagnostic	YES
Q	Oracle lower bound	44.6			oracle	YES
R	Bootstrap CI (lower_trim_20)	[33.4, 82.8]			bootstrap 95% CI	YES

The most important comparison is not Row O against every other row, but the ranking reversal under a controlled change in tag-anchor range bias. Under scalar p50, V4 wins only narrowly at 61.5 mm versus V5 at 63.5 mm. Under lower_trim_20 with Huber30, V5 reaches 44.5 mm and the best V4 raw-frame geometry row is 53.9 mm. That reversal is the evidence for delay–layout–NLOS coupling.

8 Updated Claim Matrix

The V3 report reorganizes the claim matrix around delay–layout coupling. The campaign-level counts are 10 Level A claims, 10 Level B claims, 5 Level C claims, and 2 Level D claims after removing the mixture-model ranking as a paper-level scientific claim and downgrading the lower-quantile method claim to a caveated engineering result. The full claim register remains in `FULL_V5_experimental_report/tables/claim_evidence_matrix_v2.csv`; the V3 narrative is the controlling interpretation where the prose and the raw CSV differ.

Table 8: New or revised claims introduced by the raw-frame and anchor lower-trim work.

ID	Level	Recommended wording
26	B	Lower-quantile tag-anchor aggregation improves static median and slightly improves P95 under scalar convention when enough frames are accumulated, but the absolute P95 remains high.
27	A	Inter-anchor raw ranges are nearly symmetric; lower-tail aggregation is inappropriate for anchor self-calibration here.
28	B	The e_i=0 or low-e_i V5 variant is slightly better for the raw-frame static estimator, but the bootstrap evidence is not decisive.
29	A	When tag-anchor positive bias is reduced, V4's cancellation advantage disappears and V5 becomes the better geometry.
30	A	Active vertical-geometry injection by tilted arm rotation does not reduce the vertical anchor error (dose–response slope +0.03 mm/deg, both solver models); since the cuboid layout's vertical observability is already good, this confirms the residual vertical bias is a systematic per-device delay effect, not a quantity reducible by richer vertical calibration data.

The practical wording should stay conservative. Lower_trim_20 is a static batch extractor for this campaign, not a CIR-free NLOS detector. V5 transfer to new rooms remains untested. ROTO still has a dynamic floor around 101.5 mm under the conservative BEST-FIT-ALIGNED convention. The central paper claim is the mechanism: a physically wrong self-calibration can win in the same environment because its scale error cancels structured positive range bias.

9 Engineering Recommendations

For anchor self-calibration, use p50/median inter-anchor ranges in Erlangen-style captures. The inter-anchor distributions are nearly symmetric and lower-tail aggregation worsened the blind replay. The V5 common-mode family should be reported with its e_i setting. The e_i=0 and e_reg5 variants are worth testing in the next campaign, but e_reg20 should not be silently replaced on the basis of a 24-position bootstrap with $P(\text{new wins}) = 0.659$.

For static tag solving, keep raw per-frame ranges and test lower-quantile aggregation when enough frames per link are accumulated. In this campaign, the useful row is "V5 p50 anchors, lower_trim_20 tag ranges, Huber30, D_tag LOO." The D_tag value is estimator-specific: the p50 V5 LOO value is 49.621 mm, whereas the lower-quantile V5/Huber30 row calibrates around a much smaller effective tag delay. These numbers should not be mixed.

For dynamic operation, do not transfer the static lower-tail result directly to one-frame ROTO solves. ROTO remains a separate limitation, and the current conservative headline is 101.5 mm BEST-FIT-ALIGNED. The alignment audit and time-offset sweep should be reported as diagnostics, not as a solved dynamic-tracking pipeline.

For firmware and capture design, the priority is to keep saving raw per-frame ranges. The raw data already made the ranking-flip experiment possible. A future quasi-static mode can accumulate a short histogram only when IMU and range stability indicate that the tag is not moving, then compare p50 and lower-quantile solves within that window.

10 Open Questions

The first open question is tail reduction. The lower-quantile row improves the static median and improves P95 relative to scalar V5 p50, but P95 remains around 164 mm. The next analysis should identify which positions and anchors cause this tail and whether per-anchor stability rules can reduce it without giving back the median gain.

The second open question is external validation. The current LOO and bootstrap results are still within one room and one 24-position campaign. A second-room run with a frozen pipeline should repeat p50 V4, p50 V5, lower-quantile V5, p50/e_i=0 anchors, and the same hard validation splits.

The third open question is D_tag self-calibration without known positions. The present D_tag values depend on Vicon-supported static folds or known-position residuals. Deployment needs a small-calibration procedure that can estimate the appropriate device- and estimator-specific D_tag without optical ground truth.

The fourth open question is dynamic transfer. Static lower-quantile aggregation requires a histogram, while ROTO has single-frame measurements. The next dynamic work should test quasi-static sliding windows, hardware time synchronization, and physical tag-orientation sweeps before claiming a real-time dynamic improvement.